

Forms of Energy

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Energy is the ability to cause change. Forms include **mechanical** (moving things), **light**, **sound**, **thermal** (heat), and **electrical**. Energy can change from one form to another. A lamp changes electrical energy into light and heat.

Round 1 — Retrieve It

1. What does energy let you do? Finish this sentence: Energy is the ability to...

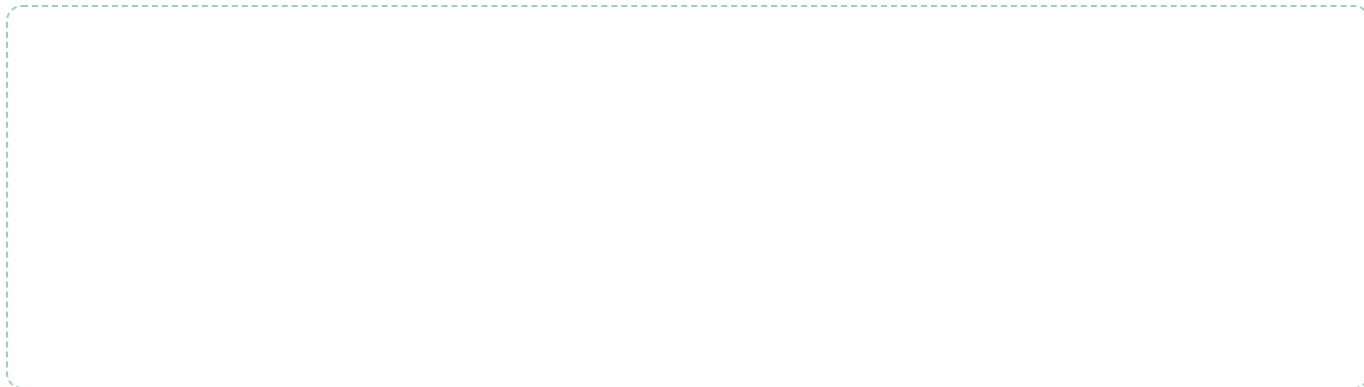
2. Name two forms of energy.

3. Which form of energy do moving objects have?

4. When you turn on a lamp, what two forms of energy come out of it?

Round 2 — Sketch It

Draw a lamp that is turned on. Use labels or arrows to show the electrical energy going in and the light and heat energy coming out.



Forms of Energy — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A radio is playing music. Which form of energy are you hearing, and where does the electrical energy change into it?

2. A person rides a bicycle down the street. Which form of energy does the moving bike have?

3. A toaster is warming a slice of bread. Name the form of energy going in and the form doing the work on the bread.

Round 4 — Reason from Evidence

A student says, “A flashlight only makes light, nothing else.” After the flashlight is left on for a while, the front of it feels warm. What does that tell you about the energy changes happening inside the flashlight?

Forms of Energy — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain in your own words what it means to say “energy can change from one form to another.” Give one example.

C**Connect**

Connect one form of energy to something you used today. Name the form and tell what change it caused.

E**Extend**

Pick one device in your home. List the form of energy that goes in and every form that comes out.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Forms of Energy

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. ...to cause change (to make things happen or move).
2. Any two: mechanical (moving things), light, sound, thermal/heat, electrical.
3. Mechanical energy (energy of motion).
4. Light energy and thermal (heat) energy.

Sketch It — look for

Electrical energy shown going in (from the plug/wire) and both light and heat energy shown coming out of the lamp.

Apply It

1. Sound energy; the electrical energy changes into sound inside the radio's speaker.
2. Mechanical energy (the energy of the moving bike).
3. Electrical energy goes in; thermal (heat) energy warms and toasts the bread.

Reason from Evidence — look for

The warmth is evidence that the flashlight makes more than light. Electrical energy changes into both light energy and thermal (heat) energy, so the student's claim is not fully correct.

ACE — Extend look-for

Device correctly names the input form and the output form(s), for example: TV — electrical energy in; light and sound out.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.